PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

GSD CarPass Depot App

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

As the number of faculty members, administrative staff, and students who own and use vehicles continues to increase, educational institutions face growing challenges in managing parking facilities effectively. The rising volume of vehicles entering and exiting campus premises can lead to congestion, parking violations, and difficulties in monitoring vehicle movements. During peak hours, the manual checking of car passes at campus entry and exit points often causes traffic buildup, resulting in longer waiting times for vehicle owners and an increased workload for security personnel. These issues not only affect the convenience of campus users but also raise concerns regarding safety, security, and the efficient utilization of available parking spaces.

Furthermore, when parking-related issues arise, such as illegal parking or vehicles obstructing designated areas, guards and General Services Department (GSD) staff often face challenges in promptly contacting vehicle owners. The lack of an efficient communication and monitoring system can delay the resolution of parking violations and hinder the enforcement of campus parking regulations. As a result, the overall management, safety, and accessibility of campus parking facilities may be negatively affected, emphasizing the need for a more efficient GSD CarPass Depot App.

At the University of Iloilo, the General Services Department (GSD) plays a crucial role in maintaining the efficiency, safety, and functionality of campus operations. The department is responsible for a wide range of services, including

building maintenance, campus safety and security, management of lost and found items, and the implementation of vehicle entry and parking policies. Through these responsibilities, the GSD helps ensure that the university environment remains organized, secure, and conducive to learning and daily operations.

One of the GSD's key responsibilities is managing vehicle entries and parking allocations within the university. Currently, the university relies on a traditional manual car pass system, requiring applicants to physically visit multiple offices and complete paper-based procedures to obtain vehicle permits. While this process has been functional in the past, the growing number of vehicles entering the campus has highlighted several challenges, including longer processing times, increased administrative workload, and traffic congestion during peak hours due to manual sticker verification at the gates. In addition, the absence of a centralized digital repository for vehicle records makes it difficult to monitor real-time campus occupancy, verify authorized vehicles efficiently, and maintain accurate records for administrative purposes.

As institutions continue to adopt digital technologies to improve operational efficiency, reliance on physical stickers, paper-based records, and manual verification procedures has become increasingly ineffective in addressing the demands of modern campus management. These limitations can slow administrative processes and reduce the effectiveness of vehicle monitoring and parking enforcement.

Moreover, a well-managed vehicle access system is essential to the smooth daily operations of the university. The timely arrival and departure of faculty members, staff, and students are critical for maintaining academic schedules and

administrative activities. Delays resulting from inefficient parking and vehicle verification procedures can negatively affect productivity and disrupt institutional operations. Therefore, adopting a technology-driven and centralized vehicle management system can significantly improve operational efficiency while providing a safer, more secure, and more convenient experience for all faculty members, staff, and students.

1.2 Statement of the Problem

The University of Iloilo's General Services Department (GSD) currently relies on a manual, paper-based GSD CarPass Depot App that has proven increasingly inadequate in addressing the demands of a growing campus population. The absence of a centralized and technology-driven solution has resulted in a range of operational inefficiencies that affect not only the administrative functions of the GSD but also the daily experiences of faculty members, administrative staff, and students who own and use vehicles within the campus. As the volume of registered vehicles continues to rise, the limitations of the existing system have become more pronounced, manifesting in processing delays, security concerns, traffic congestion, and difficulties in enforcing campus parking regulations. The persistence of these challenges underscores the critical need for a modernized GSD CarPass Depot App that can effectively respond to the evolving needs of the university community. Specifically, this study seeks to address the following problems:

Inefficient Car Pass Application Process — The existing system requires applicants to physically visit multiple offices and complete paper-based procedures, resulting in longer processing times and a heavier administrative workload for GSD personnel.

Traffic Congestion at Entry and Exit Points — Manual sticker verification during peak hours causes significant traffic buildup at campus gates, leading to extended waiting times for vehicle owners and increased burden on security personnel.

Lack of a Centralized Digital Vehicle Repository — The absence of a unified digital system makes it difficult to monitor real-time campus parking occupancy, efficiently verify authorized vehicles, and maintain accurate and up-to-date vehicle records for administrative use.

Inadequate Enforcement of Campus Parking Regulations — Without a technology-driven monitoring system, enforcing parking policies and addressing violations in a timely and systematic manner remains a persistent challenge for the GSD.

Disruption to Academic and Administrative Operations — Delays caused by inefficient vehicle verification and parking management negatively affect the punctuality and productivity of faculty, staff, and students, ultimately disrupting the smooth daily operations of the university.

1.3 Objectives of the Study

1.3.1 General Objective

This study aims to design, develop, and implement a centralized, technology-driven and Parking Management System for the University of Iloilo's General Services Department (GSD) that will comprehensively address the growing inefficiencies and operational challenges associated with the current manual,

paper-based approach to vehicle access control and parking administration. Recognizing the critical role that an efficient vehicle management system plays in sustaining the smooth and uninterrupted flow of daily academic and administrative operations, the study seeks to harness the capabilities of modern digital technology to transform and modernize the way the university handles vehicle entry, parking allocation, permit processing, and regulation enforcement within its campus premises. By transitioning to a fully digital system, the study intends to streamline the car pass application and approval process, establish a centralized digital repository for all vehicle-related records, and introduce a faster and more reliable vehicle verification mechanism that minimizes traffic congestion at campus entry and exit points during peak hours, thereby reducing the workload of security personnel and improving the overall flow of vehicle traffic within and around the university.

1.3.2 Specific Objectives

Specifically, the study aims to:

- **Develop a QR Code-Based Digital Car Pass Generation System** — Design and implement a digital car pass system that allows GSD staff to generate, manage, and issue QR code-based vehicle permits containing key vehicle information such as the plate number, owner name, owner email, and validity period, effectively replacing the existing manual, paper-based sticker system.

- **Implement an Automated QR Code Distribution via Email** — Enable the system to automatically deliver generated QR codes directly to vehicle owners through email, eliminating the need for applicants to physically visit multiple offices to claim their vehicle permits and reducing the administrative workload of GSD personnel.

- **Design a QR Code Scanning and Verification Mechanism for Security Guards** — Develop a scanning interface that allows security guards to quickly verify the validity of vehicle QR codes at campus entry and exit points, automatically detecting the status of each QR code as accepted, expired, or revoked, and logging each scan in real time to minimize manual checking and reduce traffic congestion during peak hours.

- **Establish a Centralized Digital Repository for Vehicle Records and Entry Logs** — Create a unified digital database that stores all QR code records, vehicle information, and entry and exit history, enabling authorized personnel to efficiently retrieve, monitor, and manage vehicle data and campus parking occupancy at any given time.

- **Provide an Administrative Dashboard for Real-Time Parking Monitoring and Reporting** — Develop an admin dashboard that displays real-time data on parking slot availability, total vehicle entries and exits, active QR codes, and recent activity logs, enabling the GSD administration to make informed decisions regarding parking management and space utilization.

- **Enable Role-Based Access Control for System Users** — Implement a secure, role-based access control mechanism that assigns distinct permissions and functionalities to three user roles — Admin, Staff, and Guard — ensuring that each user can only access the features and data relevant to their designated responsibilities within the system.

1.4 Scope and Delimitation

Scope

This study covers the development and implementation of the GSD CarPass Depot App, a web-based application designed exclusively for the General Services Department (GSD) of the University of Iloilo. The system aims to modernize and streamline the department's vehicle access control and parking management operations through the use of QR code-based vehicle permits and digital record management.

The system accommodates three user roles with distinct functionalities. The Admin manages system user accounts, including the creation, viewing, and deletion of Admin, Staff, and Guard accounts; configures and adjusts total parking slot capacity; monitors real-time parking occupancy; views vehicle entry and exit activity logs; and generates reports through a comprehensive administrative dashboard. The Staff generates and issues QR code-based vehicle permits, manages vehicle owner information such as plate numbers and email addresses, delivers QR codes to vehicle owners via automated email, renews and reactivates expired QR code permits, and views the complete history of all generated QR codes. The Guard scans and verifies QR codes at campus entry and exit points using a dedicated scanning interface, logs each scan transaction with its corresponding status (accepted, expired, or revoked), and views personal scan history.

The GSD CarPass Depot App covers the generation and issuance of QR code-based vehicle permits containing essential information, including the vehicle plate number, owner name, owner email address, and permit validity period. An automated email delivery mechanism transmits the approved QR code-based vehicle permit directly to the registered email address of the vehicle owner upon generation by GSD staff. The system also supports the renewal and reactivation of expired QR code permits, allowing GSD staff to extend the validity of existing vehicle passes without generating entirely new QR codes.

The GSD CarPass Depot App provides real-time monitoring of parking slot availability and occupancy, automatically updating parking status based on QR code scan

data recorded at campus entry and exit points. A centralized digital repository maintains all vehicle records, QR code permits, ownership information, and vehicle entry and exit history, accessible only to authorized GSD personnel. Activity logs and reports documenting vehicle entry and exit transactions, QR code scan results, and overall parking utilization are generated by the system and are accessible to the Admin for administrative and reporting purposes. Profile management is also supported, allowing system users to update their account email address and password credentials.

The GSD CarPass Depot App is built using the Flask Python web framework for the backend, with an HTML, CSS, and JavaScript frontend, and a MySQL relational database for data storage and management. It does not have a dedicated mobile application; however, the QR code scanning interface used by security guards may be accessed through a mobile web browser, as the scanning feature utilizes the device's camera for QR code reading. The system requires a stable internet connection for all core functionalities, including QR code generation, email delivery, scanning and verification, and real-time parking monitoring. A secure authentication mechanism restricts access to authorized users only, with role-based permissions ensuring that each user can access only the features relevant to their designated role. The GSD CarPass Depot App is designed exclusively for deployment and use within the operational context of the University of Iloilo's General Services Department and is not intended for use by vehicle owners or the general public.

Delimitation

The study is bounded by the following limitations in terms of the scope and coverage of the **GSD CarPass Depot App**.

The system does not cover the online submission of car pass applications by vehicle owners. Applicants are still required to manually accomplish application forms, pay the corresponding car pass fees, and personally submit their documentary requirements to the GSD office prior to QR code permit generation. Accordingly, the system does not include an

automated payment processing feature or online collection of car pass fees, as the collection, verification, and recording of fees remain a manual responsibility of the GSD and are outside the scope of the system.

The system does not incorporate physical detection of vehicles using sensors, cameras, or any hardware-based vehicle detection technology for monitoring actual parking space utilization within the campus premises. Parking occupancy tracking is based solely on QR code scan data recorded at campus entry and exit points. As such, the system does not automatically detect, flag, or report parking violations, unauthorized parking, or vehicles occupying restricted areas. Physical monitoring and enforcement by security personnel remain necessary for these concerns. Real-time GPS tracking or location monitoring of vehicles within the campus premises is likewise not supported.

The system does not include a notification or alert system directed to vehicle owners regarding the status of their QR code permits, such as upcoming expiration reminders, renewal notices, or parking violation alerts. The system also does not support integration with third-party systems such as the university's enrollment system, human resource system, or any other existing institutional database. Furthermore, the system does not cover the generation of printed physical car pass stickers or any other form of physical vehicle permit, nor does it cover the management of pedestrian access, visitor passes, or non-motorized vehicles such as bicycles within the campus premises.

The system is accessible only via a web browser and does not include a dedicated mobile application for any of the three user roles on Android or iOS platforms. All core functionalities of the system, including QR code generation, email delivery, scanning and verification, and real-time parking monitoring, require a stable internet connection, as offline or local network operation is not supported. Finally, the **GSD CarPass Depot App** is designed exclusively for the operational requirements of the University of Iloilo's General Services Department and is not intended for deployment in other institutions without

significant modifications to accommodate different parking management structures and policies.

1.5 Significance of the Study

Significance of the Study

This study holds considerable significance as it addresses the real and pressing operational challenges faced by the University of Iloilo's General Services Department in managing vehicle access and parking within the campus. The development and implementation of the **GSD CarPass Depot App** is expected to bring meaningful improvements to the operational efficiency, security, and convenience of campus vehicle management. The following are the primary beneficiaries of this study:

General Services Department (GSD). The GSD stands as the primary beneficiary of this study. The system significantly reduces the administrative burden associated with the manual processing, issuance, and distribution of vehicle permits by digitizing the generation and delivery of QR code-based car passes. With a centralized digital repository for vehicle records and a real-time dashboard for monitoring parking occupancy and activity logs, the GSD will be better equipped to manage campus vehicle access efficiently, enforce parking policies, and maintain accurate and organized records without relying on paper-based procedures.

GSD Staff. The system directly improves the day-to-day workflow of GSD staff members by providing a streamlined digital interface for generating, issuing, renewing, and managing QR code-based vehicle permits. The automated email delivery of approved car passes eliminates the need for staff to manually distribute physical permits, allowing them to dedicate more time and attention to other essential administrative responsibilities within the department.

Security Guards. Security personnel stationed at campus entry and exit points will benefit from a faster and more reliable vehicle verification mechanism. The QR code scanning interface enables guards to instantly determine the validity status of a vehicle permit as accepted, expired, or revoked, minimizing the time spent on manual sticker verification and reducing the occurrence of traffic congestion during peak hours. This also lessens the physical workload of guards and enhances the overall security of campus entry and exit operations.

Faculty Members and Administrative Staff. Faculty and administrative staff who own and use vehicles within the campus will experience a more convenient and efficient car pass issuance process. The automated delivery of QR code permits directly to their registered email addresses eliminates the need to make multiple visits to the GSD solely to claim their vehicle permits, saving time and effort while ensuring timely receipt of their approved passes. Additionally, the QR code renewal feature allows vehicle owners with expired permits to have their passes reactivated more efficiently without undergoing the full application process again.

University Administration. The University of Iloilo's administration will benefit from the improved organizational efficiency and enhanced security that the system brings to campus operations. The availability of real-time data on parking occupancy, vehicle entries and exits, and permit activity provides administrators with valuable insights for informed decision-making regarding parking infrastructure, policy enforcement, and resource allocation. The system also supports the university's broader institutional goal of adopting modern digital technologies to improve operational efficiency, enhance campus safety and security, and provide a better environment for all members of the university community.

Future Researchers and Developers. This study serves as a foundational reference and practical resource for future researchers and developers who may seek to further enhance, expand, or build upon the GSD CarPass Depot App. The system's modular architecture,

role-based access control design, and documented codebase provide a solid and scalable foundation for future development efforts. Future developers may extend the system by incorporating additional features such as online car pass application and submission, automated payment processing, hardware-based vehicle detection, mobile application development, or integration with other existing university systems. The insights, methodologies, and lessons documented in this study can likewise guide future researchers in addressing similar challenges in other institutional or organizational contexts, contributing to the broader body of knowledge in the field of campus management systems and information technology development.

1.6 Definition of Terms

Definition of Terms

The following terms are defined both conceptually and operationally to provide clarity and a common understanding of the key concepts used throughout this study:

Vehicle Access and Parking Management System. *Conceptual* — A vehicle access and parking management system refers to an integrated set of tools, technologies, and procedures designed to regulate, monitor, and control the entry, exit, and parking of vehicles within a defined area or facility, with the goal of improving security, efficiency, and space utilization (Rodrigue, 2020). *Operational* — In this study, the Vehicle Access and Parking Management System refers to the web-based application developed for the General Services Department of the University of Iloilo that manages the generation and issuance of QR code-based vehicle permits, monitors real-time parking occupancy, and records vehicle entry and exit transactions within the campus premises.

QR Code (Quick Response Code). *Conceptual* — A QR code is a two-dimensional matrix barcode that stores information in a pattern of black and white squares and can be scanned using a digital device such as a smartphone or barcode reader to quickly retrieve encoded data (Denso Wave, 2020). *Operational* — In this study, QR code refers to the unique, digitally generated code embedded in each vehicle permit issued by the GSD staff through the system, which contains the vehicle owner's information and permit validity details and is scanned by security guards at campus entry and exit points for verification purposes.

Car Pass. *Conceptual* — A car pass is an official authorization document or permit issued by a governing authority or institution that grants a vehicle owner the right to access, enter, and park within a designated area or facility (Cambridge Dictionary, 2023). *Operational* — In this study, a car pass refers to the QR code-based digital vehicle permit generated and issued by GSD staff through the system to authorized faculty members, administrative staff, and students of the University of Iloilo, granting them the right to enter and park within the campus premises.

General Services Department (GSD). *Conceptual* — A general services department is an organizational unit within an institution responsible for managing and overseeing a wide range of support and administrative services, including facilities maintenance, security, and the implementation of institutional policies and procedures (Merriam-Webster, 2022). *Operational* — In this study, the General Services Department (GSD) refers to the specific department of the University of Iloilo responsible for managing vehicle access, parking allocation, and the issuance of car pass permits within the campus, and which serves as the primary end-user and beneficiary of the developed system.

Real-Time Monitoring. *Conceptual* — Real-time monitoring refers to the continuous and immediate observation, tracking, and recording of data or activities as they occur, enabling users to access up-to-date information and respond promptly to changing conditions (Techopedia, 2021). *Operational* — In this study, real-time monitoring refers to the system's

capability to instantly update and display the current parking occupancy, vehicle entry and exit counts, and QR code scan activity on the administrative dashboard as transactions are recorded at campus entry and exit points.

Role-Based Access Control (RBAC). *Conceptual* — Role-based access control is a security mechanism that restricts system access and functionalities to users based on their assigned roles within an organization, ensuring that individuals can only perform actions and view information relevant to their designated responsibilities (Ferraiolo & Kuhn, 1992). *Operational* — In this study, role-based access control refers to the system's security feature that assigns distinct permissions and functionalities to three user roles — Admin, Staff, and Guard — preventing unauthorized access to features and data outside the scope of each user's designated role within the GSD.

Web-Based Application. *Conceptual* — A web-based application is a software program that is hosted on a remote server and accessed by users through a web browser over the internet or a local network, without requiring installation on the user's device (Laudon & Laudon, 2020). *Operational* — In this study, web-based application refers to the GSD CarPass Depot App developed for the General Services Department of the University of Iloilo, which is accessible through a web browser by authorized Admin, Staff, and Guard users to perform their respective vehicle management functions.

Parking Occupancy. *Conceptual* — Parking occupancy refers to the proportion or number of parking spaces within a facility that are currently in use at a given point in time, often used as a measure of parking demand and space utilization (Shoup, 2011). *Operational* — In this study, parking occupancy refers to the number of vehicles currently present within the campus parking area as tracked and displayed by the system's administrative dashboard, based on the QR code scan data recorded at campus entry and exit points.

Automated Email Delivery. *Conceptual* — Automated email delivery refers to the process by which a software system automatically generates and sends electronic mail messages to

designated recipients upon the occurrence of a specific trigger or event, without requiring manual intervention from the sender (Chaffey, 2019). *Operational* — In this study, automated email delivery refers to the system's feature that automatically transmits the approved QR code-based vehicle permit to the registered email address of the vehicle owner immediately upon its generation by GSD staff, eliminating the need for manual distribution of physical car passes.

QR Code Renewal. *Conceptual* — QR code renewal refers to the process of extending or updating the validity period of an existing QR code, allowing it to remain functional and authorized beyond its original expiration date (ISO/IEC 18004, 2015). *Operational* — In this study, QR code renewal refers to the system feature that enables GSD staff to update the expiry date of an existing vehicle permit QR code, reactivating expired car passes and allowing vehicle owners to continue using their previously issued QR codes without the need for a completely new permit to be generated.

Entry and Exit Logs. *Conceptual* — Entry and exit logs are systematic records that document the movement of individuals or vehicles into and out of a facility, typically including details such as the time, date, and identity of the person or vehicle involved (Black, 2019). *Operational* — In this study, entry and exit logs refer to the digital records automatically generated and stored by the system each time a vehicle's QR code is scanned at a campus gate by a security guard, capturing the scan timestamp, QR code details, vehicle plate number, and whether the transaction was an entry or exit.

Centralized Digital Repository. *Conceptual* — A centralized digital repository is a single, unified storage system that consolidates and organizes digital data and records from multiple sources, making them accessible to authorized users from one central location (Arms, 2000). *Operational* — In this study, centralized digital repository refers to the system's unified database that stores and maintains all vehicle permit records, QR code information, vehicle

owner details, and entry and exit transaction logs, which can be accessed and managed by authorized GSD personnel through the web-based system.